



Suno bhai, ye chapter waise to mathematics ka hai jo maths wale bachche 12th class mein extreme detail mein padhenge. Yahan humein utna hi padhna hai jitna physics mein use hona hai. Zaroorat se zyada PHD karne ki koshish na karein, because woh hum 12th mein karenge.

Kuch bachche jo ise pehli baar padh rahe hain, unhein starting mein yeh chapter mushkil lag sakta hai. Unse main yahi kahunga ki jitna content main yahan de raha hoon, usko 3-4 baar achche se rough copy par practice kar lein. Itne se physics mein achche se kaam chal jayega.

Scalar Quantity

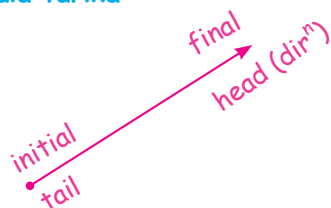
Those physical quantities which can be completely described by its magnitude only are called Scalar.

Vector Quantity

Those physical quantity which have magnitude & direction and they follow law of vector algebra.
Eg. Force, Acceleration, Momentum.

REPRESENTATION OF VECTOR

Diagram wala tarika



Length of arrow represent magnitude of vector.

SKC

Kaun si physical quantity vector hai ya scalar hai abhi hume is baat se koi lena dena nahi hai. Kyunki abhi hamari physics start nahi hui hai, so take it lightly. Vector ka surur hole hole chadhega.

Mathematically/Analytical

★ A vector $\vec{P}$ is represented by $\vec{P}$

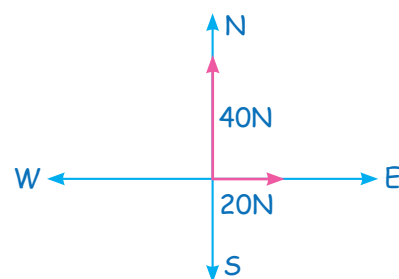
★ Force = $\vec{F}$

Q. Represent vector $\vec{P}$ & vector $\vec{Q}$

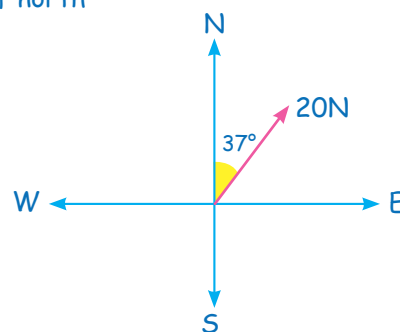
$\vec{P}$ = 20N along East

$\vec{Q}$ = 40N along north

Sol.



★ Represent $\vec{P}$ of magnitude 20N in direction of 37° east of north



TYPES OF VECTOR

Abe, achche se inki reading le lena.



Equal Vector

Two vector are said to be equal vector, if they have same magnitude, same direction and same physical quantity.



$\vec{R}$ and $\vec{P}$ are equal vector : ✗

$\vec{P}$ and $\vec{Q}$ are equal vector : ✓

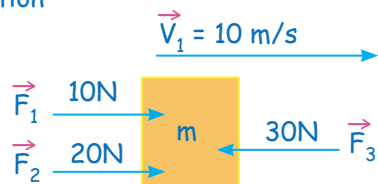
Parallel Vector →

Two vectors are said to be parallel if they have same direction.

Here, Physical quantity, alag alag ho sakti hai. All equal vectors are parallel vectors

Anti-Parallel Vector = Direction Opposite

A Block is moving with constant velocity 10 m/s along east direction



$\vec{F}_1$ & $\vec{F}_2$ are parallel vector.

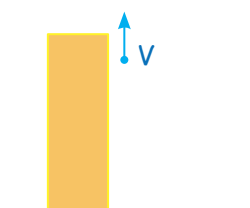
$\vec{F}_1$ & $\vec{F}_2$ are not equal vector.

$\vec{F}_1$ & $\vec{V}_1$ are parallel vector

$\vec{F}_2$ & $\vec{V}_1$ are parallel vector

$\vec{F}_1$ & $\vec{F}_3$ are anti-parallel vector

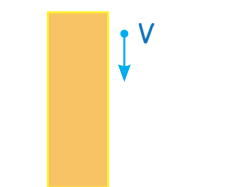
$\vec{V}_1$ & $\vec{F}_3$ are anti-parallel vector.



$\vec{V} \rightarrow$ Upar

$\vec{a} \rightarrow$ Neeche

$\vec{V}$, $\vec{a}$ are antiparallel.



$\vec{V} \rightarrow$ Neeche

$\vec{a} \rightarrow$ Neeche

$\vec{V}$, $\vec{a}$ are parallel.

Negative Vector

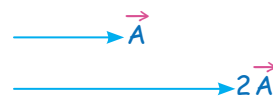
Two vectors $\vec{A}$, & $\vec{B}$, are said to be negative of each other if they have same magnitude & opposite direction.



$\vec{A} = -\vec{B}$ are negative vector to each other.

Multiplication of a Vector with a Number

★ If a vector $\vec{A}$ is multiply by a number $n > 0$. Then magnitude of vector becomes n times & direction remains same.



★ If a vector $\vec{A}$ is multiply by a number $n < 0$, then magnitude of vector becomes n times & direction become opposite.



SKC

Agar humne kisi vector ko 'n' se multiply kiya, to vector ki length 'n' times ho jaegi. Agar 'n' positive hai to direction same rahegi aur agar 'n' negative hai to direction opposite ho jaegi.



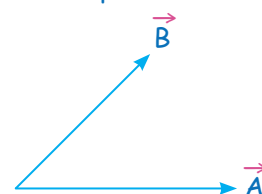
Coplanar Vector

If all vectors lie in same plane then vectors are known as co-planar.

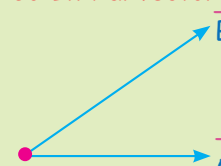
NOTE:- TWO VECTORS are always COPLANAR

Coinitial Vector

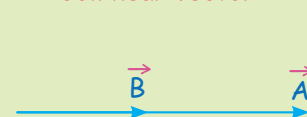
[Tail to Tail]: Two vectors are said to be coinitial if they have same initial point.



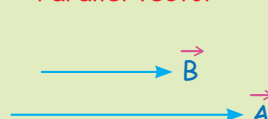
Co-Initial vector



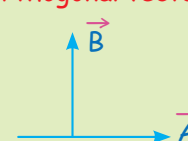
Collinear vector



Parallel vector



Orthogonal vector



Kaun si physical quantity scalar hai kaun si vector hai, ye tu yaad kar raha hai????



Haa humne RATT liye



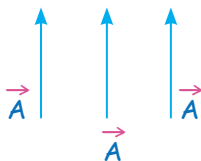
Nhi RATTNA tha

Poore do saal yahi padhna hai ki kaun si physical quantity kya hai, so is par time bilkul waste na karein aur mast rahe.



Parallel Shifting of Vector

Whenever we need, we can parallelly shift any vector without changing direction and magnitude we can say that vector will not change.



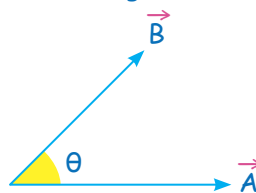
SKC



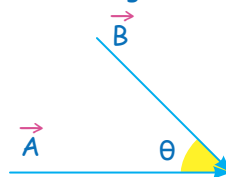
- ★ Dekh bhai, abhi hum apni zaroorat ke anusar vector ko parallelly shift kar sakte hain, jaise computer ke mouse ka cursor sign hota hai.
- ★ Agar kisi vector ki direction badli, matlab vector badal jaega.
- ★ Agar kisi vector ka magnitude badla matlab vector badal jaega.
- ★ Agar magnitude, direction same = vector same.

Angle Between Two Vectors

★ It is the angle b/w their tail.



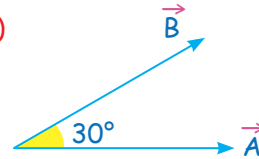
★ It is the angle b/w their head.



$[0 < \theta < 180]$ $[\theta \rightarrow \text{angle b/w } \vec{A} \text{ \& } \vec{B}]$

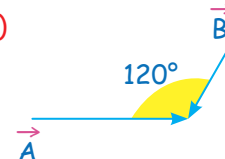
Q. Find angle b/w $\vec{A}$ & $\vec{B}$

(a)



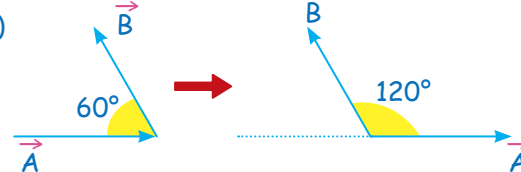
Angle b/w vectors = 30°

(b)



Angle b/w vectors = 120°

(c)

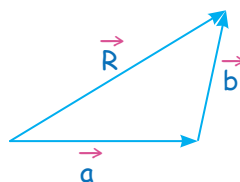


Angle b/w vectors = $180 - 60^\circ = 120^\circ$

ADDITION OF 2 VECTORS

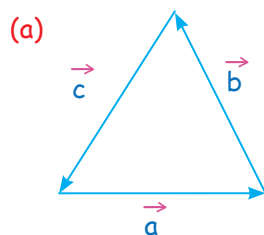
Triangle Rule of Vector Addition

If 2 vectors represent two sides of a triangle in same sense (head of first vector coincide with tail of other vector) resultant is 3rd side of triangle in opposite sense.



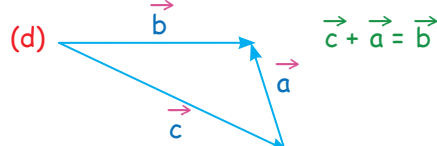
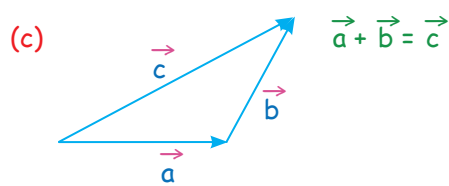
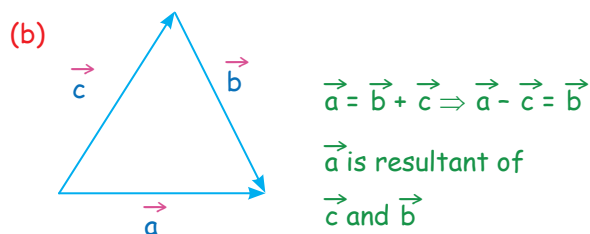
$$\vec{a} + \vec{b} = \vec{R}$$

Q. Write the eqn for following fig. using triangle of vector addition.



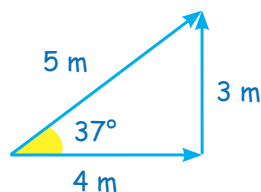
Closed loop same sense

$\vec{a} + \vec{c} + \vec{b} = 0$ iska matlab teeno vector ka sum/addition/resultant 0 aa gaya.



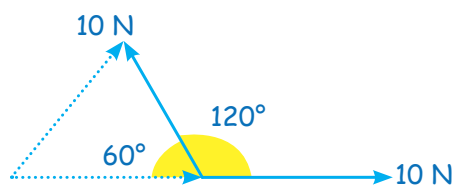
Q. A man moves 4m in east direction and then 3m in north direction. Find the resultant displacement of man.

Sol.



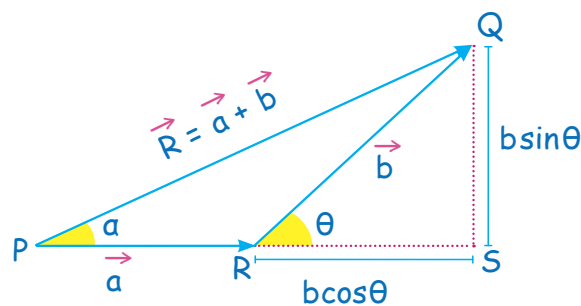
5 m in 37° north of east OR 5 m in 53° East of north.

Q. Find the resultant of the forces shown.



Sol. Resultant 10 in 60° north of east.

ANALYTICAL METHOD TO FIND RESULTANT AND ITS DIRECTION (OF 2 VECTORS)



$$|\vec{a}| = a, |\vec{b}| = b, |\vec{R}| = |\vec{a} + \vec{b}| = R$$

In ΔRQS

$$\sin \theta = QS/b \Rightarrow QS = b \sin \theta$$

$$\cos \theta = RS/b \Rightarrow RS = b \cos \theta$$

In ΔPQS

$$(PQ)^2 = (PR + RS)^2 + QS^2$$

$$R^2 = (a + b \cos \theta)^2 + (b \sin \theta)^2$$

$$R^2 = a^2 + b^2(\cos^2 \theta + \sin^2 \theta) + 2ab \cos \theta$$

$$R = \sqrt{a^2 + b^2 + 2ab \cos \theta}$$

$$\tan \alpha = \frac{b \sin \theta}{a + b \cos \theta}$$

R = Magnitude of resultant.

α = Angle b/w $\vec{R}$ & $\vec{a}$

Dekh bhai, kaam ki baat yeh hai ki do vector $\vec{a}$ and $\vec{b}$ ke resultant ka magnitude $R = \sqrt{a^2 + b^2 + 2ab \cos \theta}$ hota hai jo $\vec{a}$ ke saath angle α banata hai.

Where $\tan \alpha = \frac{b \sin \theta}{a + b \cos \theta}$ ye formula

galti se bhi mat bhoolna, warna pitaai hogi agar resultant R ne $\vec{b}$ ke saath

angle β banaya to $\tan \beta = \frac{a \sin \theta}{b + a \cos \theta}$



Q. Two vector of magnitude 10N each gives a resultant of magnitude $10\sqrt{3}$ N. Find angle b/w both vector.

Sol. $R = 10\sqrt{3}$ N

$$R = \sqrt{10^2 + 10^2 + 200\cos\theta}$$

$$300 = 200 + 200\cos\theta \Rightarrow \theta = 60^\circ$$

Q. Two forces $\vec{A}$ and $\vec{B}$ of magnitude 10 N and 20 N are acting on a block having 60° angle between them. Find the magnitude of resultant of them

Sol. $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$$= \sqrt{100 + 400 + 200} = \sqrt{700}$$

Angle made by resultant of $\vec{A}$ & $\vec{B}$ with $\vec{A}$.

$$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta} = \frac{20 \times \frac{\sqrt{3}}{2}}{10 + 20 \times \frac{1}{2}} = \frac{\sqrt{3}}{2}$$

SPECIAL CASES

1 $\theta = 0^\circ$ (Parallel Vectors)



$$\vec{R} = \vec{a} + \vec{b}$$

$$R = \sqrt{a^2 + b^2 + 2ab} = a + b$$

$$R_{\max} = a + b = |\vec{a}| + |\vec{b}|$$

2 $\theta = 180^\circ$ (Antiparallel Vectors)



$$\vec{R} = \vec{a} + \vec{b}$$

$$R = \sqrt{a^2 + b^2 - 2ab} = |a - b|$$

$$R_{\min} = |a - b| = ||\vec{a}| - |\vec{b}||$$

$$|a - b| \leq R \leq (a + b)$$

$$R_{\max} = a + b \quad R_{\min} = |a - b|$$

3 $\theta = \pi/2$ or 90°

$$R = \sqrt{a^2 + b^2}$$

NOTE: Value of θ increases from 0° to 180° then magnitude of resultant will decrease.

Vector



Vector ki understanding badhane ke liye neeche kuch important ques. attach kar raha hun. Bhale hi aapko kuch ques. easy lage, phir bhi aapko sare ques. in the last solve karne hain.



Q. Max and min magnitude of resultant of two forces are 7 N and 1 N respectively. Find the resultant of 2 forces when it act orthogonally.

Sol. $F_1 + F_2 = 7$ N;

$$F_1 - F_2 = 1$$
 N

$$\Rightarrow F_1 = 4$$
 N; $F_2 = 3$ N

When F_1 and F_2 are perpendicular to each other

$$\text{then } R = \sqrt{3^2 + 4^2} = 5$$

Q. Two vector A & B with same magnitude x. Find magnitude of resultant of A and B. $\theta = 60^\circ$.

Sol. $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$$= \sqrt{2x^2 + 2x^2 \cos 60^\circ}$$

$$= \sqrt{2x^2 + 2x^2 \frac{1}{2}} = \sqrt{3x^2} = x\sqrt{3}$$

Q. In above question if $A = x$, $B = x$, $\theta = 120^\circ$, $R = ?$

Sol. $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$$= \sqrt{2x^2 + 2x^2 \left(-\frac{1}{2}\right)}$$

$$= \sqrt{x^2} = x$$

Q. If two vector $\vec{A}$ and $\vec{B}$ of magnitude 10 N and 6 N are at an angle 60° . If $\vec{B}$ become twice to its initial value & added to $\vec{A}$. Find magnitude of resultant of A & B after $\vec{B}$ change.

Sol. $B = 2 \times 6 = 12$ N

$$A = 10$$
 N

$$C = \sqrt{100 + 144 + 2 \times 10 \times 12 \times \cos 60} = \sqrt{364}$$

Q. Magnitude of $\vec{A}$ is 8 N.

Magnitude of $\vec{B}$ is 6 N.

Which of the following can be magnitude of $\vec{A} + \vec{B}$

Abe, θ to 60° hi rahega.

- (a) 10 N (b) 22 N (c) 48 N
 (d) 2 N (e) 2.001 N (f) 1.99 N
 (g) 14.1 N (h) 13.999 N

Sol. $C_{\max} = A + B = 14$
 $C_{\min} = A - B = 2$
 $2 \leq C \leq 14$

Ans. (a), (d), (e), (h)

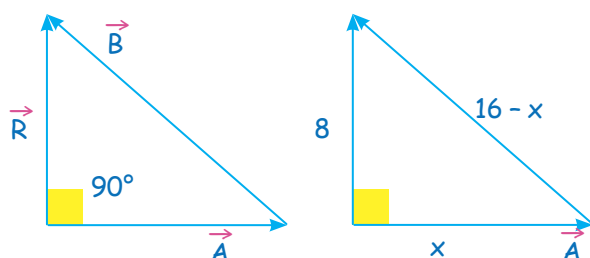
Q. Magnitude of resultant of $\vec{A}$ and $\vec{B}$ is 5 unit where magnitude of $\vec{A}$ is $5\sqrt{3}$ unit and magnitude of $\vec{B} = 5$ unit. Find angle between A and B.

Sol. $\vec{C} = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
 $5^2 = 75 + 25 + 50\sqrt{3}\cos\theta$
 $\frac{-\sqrt{3}}{2} = \cos\theta \Rightarrow \theta = 150^\circ$

Q. Sum of the magnitude of $\vec{A}$ and $\vec{B}$ is 16 N. Magnitude of resultant of vector $\vec{A}$ and $\vec{B}$ is 8 N. When resultant is perpendicular to the $\vec{A}$. Find magnitude of $\vec{A}$ and $\vec{B}$.

Sol. $A + B = 16$ N
 Magnitude of $\vec{R} = 8$ N
 $\alpha = 90^\circ$
 $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$, $\tan \alpha = \frac{B\sin\theta}{A + B\cos\theta}$
 $\tan 90^\circ = \frac{B\sin\theta}{A + B\cos\theta} \Rightarrow A + B\cos\theta = 0$
 $B\cos\theta = -A$
 $64 = A^2 + B^2 + 2AB\cos\theta$
 $64 = A^2 + B^2 - 2A^2$
 $64 = B^2 - A^2 = (A + B)(B - A)$
 $64 = 16(B - A)$
 $4 = (B - A)$
 $B - A = 4$
 $B + A = 16$
 $\frac{2B = 20}{B = 10}$
 $A = 6$

Method-2:



By using Pythagoras theorem.

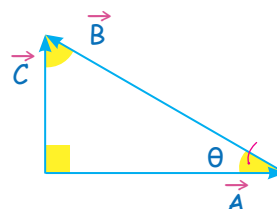
$$(16 - x)^2 = 8^2 + x^2$$

$$256 + x^2 - 32x = 64 + x^2$$

$$256 - 32x = 64$$

$$\Rightarrow 6 = x \text{ \& } 16 - x = 10$$

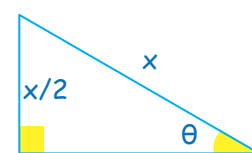
Q. Resultant of $\vec{A}$ and $\vec{B}$ is $\perp$ to $\vec{A}$ and its magnitude is equal to half of magnitude of $\vec{B}$. Find the angle between $\vec{A}$ and $\vec{B}$.



Sol. $\vec{A} + \vec{B} = \vec{C}$

$$\sin\theta = \frac{x/2}{x} = \frac{1}{2}$$

$$\theta = 30^\circ$$



Angle between $\vec{A}$ & $\vec{B} = 180^\circ - 30^\circ = 150^\circ$

Angle between $\vec{B}$ and $\vec{C} = 60^\circ$

Angle between $\vec{A}$ and $\vec{C} = 90^\circ$

3. Two vector $\vec{A}$ and $\vec{B}$ have same magnitude 'a' and resultant has magnitude R. Now $\vec{B}$ is doubled and added to $\vec{A}$ and now new resultant become $a\sqrt{3}$. Find angle b/w $\vec{A}$ and $\vec{B}$.

Sol. $\vec{A} + \vec{B} = \vec{R}$

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$R_{\text{old}} = \sqrt{2a^2 + 2a^2\cos\theta}$$

$$|2\vec{B} + \vec{A}| = a\sqrt{3}$$

$$R_{\text{new}} = \sqrt{a^2 + (2a)^2 + 2(a)(2a)\cos\theta}$$

$$a\sqrt{3} = \sqrt{a^2 + 4a^2 + 4a^2\cos\theta}$$

$$3a^2 = 5a^2 + 4a^2\cos\theta$$

$$-2a^2 = 4a^2\cos\theta$$

$$\frac{-2}{4} = \frac{-1}{2} = \cos\theta$$

$$\theta = 120^\circ$$



Kaam ka Dabba

★ If $\vec{A} + \vec{B} = \vec{R}$

Magnitude of $\vec{R} = R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

θ = angle between A & B

★ $\tan\alpha = \frac{b\sin\theta}{a + b\cos\theta}$ (Kaise aaya Not Important)

(α is the angle made by $\vec{R}$ with $\vec{A}$)

★ $R = |\vec{A} + \vec{B}|$ = magnitude of resultant of $\vec{A}$ and $\vec{B}$.

★ $|A - B| \leq R \leq (A + B)$

★ Do vector ka resultant max tab hoga jab unke beech angle 0° hoga. Aur min tab hoga jab unke beech angle 180° hoga.

If $\theta = 0 \Rightarrow R = A + B = R_{\max}$

If $\theta = 180 \Rightarrow R = |A - B| = R_{\min}$ = [Bada-Chhota]

If $\theta = 90 \Rightarrow R = \sqrt{A^2 + B^2}$

★ If $|\vec{A}| = |\vec{B}| = A$ (let)

$\theta = 0 \rightarrow R = 2A$

$\theta = 60^\circ \rightarrow R = A\sqrt{3}$

$\theta = 90^\circ \rightarrow R = A\sqrt{2}$

$\theta = 120^\circ \rightarrow R = A$

$\theta = 180^\circ \rightarrow R = 0$

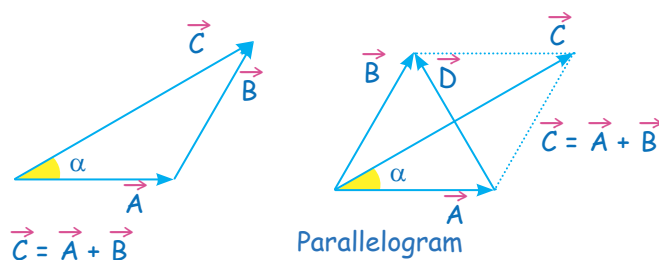
$R = 2A \cos \theta/2$

★ 2 equal magnitude wale vector ka resultant unake bicho-bich (along angle bi-sector) nikalega.

2 equal magnitude ke vector agar 60° par hai to unka resultant $\sqrt{3}$ times i.e $A\sqrt{3}$ hoga aur vo 120° par hai to unka resultant A hoga.

PARALLELOGRAM LAW OF VECTOR ADDITION

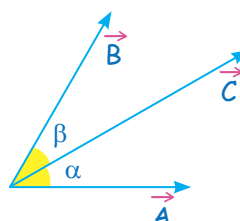
If two cointial vectors are given then resultant of these two vectors are given by diagonal of parallelogram (II gm) made from 2 given vector by shifting them parallel to their cointial vector & other diagonal of the II gm gives difference of vectors.



Parallelogram

$\vec{A} + \vec{D} = \vec{B}$

$\vec{D} = \vec{B} - \vec{A}$



If $A > B$, $\alpha < \beta$

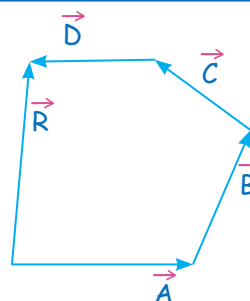
If $A < B$, $\alpha > \beta$

[Jo Vector bhari hoga Resultant us taraf jhukega.]

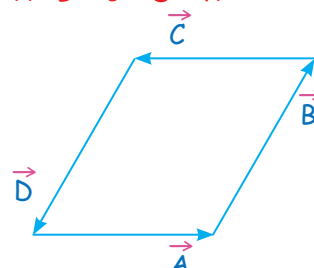
POLYGON-LAW

SKC

Resultant vector ko draw karne ke liye pahle ki Tail ko Last ke Head se Join Krdo



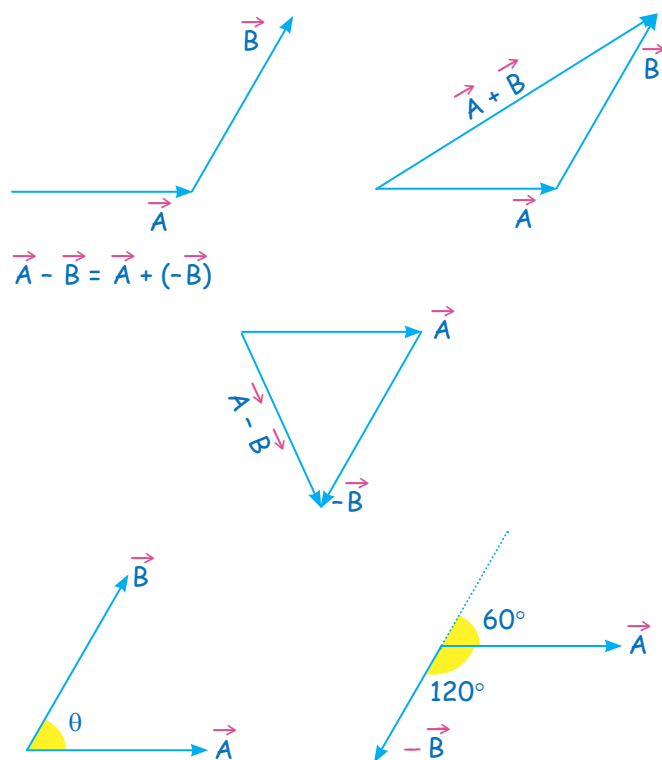
$\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{R}$



$\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{O}$ [zero vector]

Dirxn of this zero vector is arbitrary. [Jo Man Kre]
It is a vector where magnitude is zero.

SUBTRACTION OF TWO VECTOR



$\vec{A}$ aur $\vec{B}$ ke bich agar angle θ hai to $\vec{A}$ me aur $-\vec{B}$ ke bich $180-\theta$ angle hoga

If angle between $\vec{A}$ & $\vec{B}$ is θ ,

Then, $\vec{C} = \vec{A} + \vec{B}$

$$C = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta}$$

SKC

Poori Physics mein kisi vector ki physical quantity mein (-) sign ka matlab yeh hoga ki jidhar aapne positive direction maani hai, uske opposite physical quantity ki direction hai.

[(-) matlab direction se lena dena]



$$\vec{D} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

↳ Magnitude = ?

angle between $\vec{A}$ & $-\vec{B}$ will be $180 - \theta$.

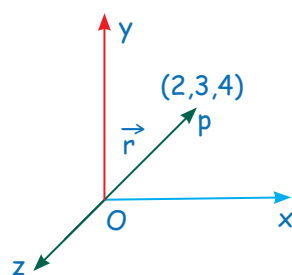
$$D = \sqrt{A^2 + B^2 + 2AB\cos(180 - \theta)}$$

$$= \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

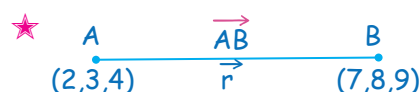
$$\tan\alpha = \frac{B\sin(180 - \theta)}{A + B\cos(180 - \theta)} = \frac{B\sin\theta}{A - B\cos\theta}$$

POSITION VECTOR

A vector representing location or position of a point in space w.r.t origin is called position vector.



Position vector of P w.r.t origin = $2\hat{i} + 3\hat{j} + 4\hat{k}$

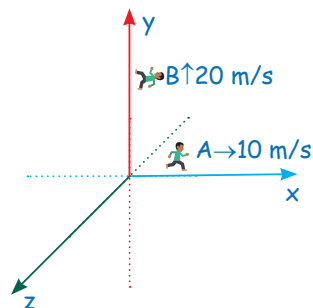


$\vec{AB} = \text{P.V of B} - \text{P.V of A}$

$$= (7-2)\hat{i} + (8-3)\hat{j} + (9-4)\hat{k} = 5\hat{i} + 5\hat{j} + 5\hat{k}$$

Q. A particle A is moving with speed 10 m/s along + x-axis and another particle B is moving along y-axis with speed 20 m/s. Find their velocity.

Sol.



→ Velocity of A = $10\hat{i}$

→ Velocity of B = $20\hat{j}$

MAGNITUDE OF A VECTOR

Q. $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$

Sol. Magnitude of $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$

Q. $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $|\vec{A}| = ?$

Sol. Magnitude of A

$$= \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

Q. $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$, $|\vec{A}| = ?$

Sol. Magnitude of A

$$= \sqrt{2^2 + 3^2 + (-4)^2} = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$$

Q. $\vec{A} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$. Find magnitude of $\vec{A}$.

Sol. $|\vec{A}| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1$

Q. $\vec{A} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}$, $|\vec{A}| = ?$

Sol. $|\vec{A}| = \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2} = 1$

UNIT VECTOR

hai to vector hi $\hat{A}$ vector whose magnitude is 1. It represent dirx^n .
Andhe ki laathi.

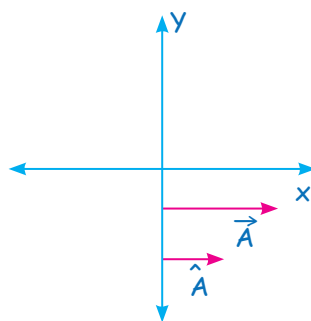
it is used to give dirx^n and is unitless & dimensionless

$$\vec{A} = |\vec{A}| \text{dirx}^n$$

$$\vec{A} = |\vec{A}| \hat{A}$$

unit vector

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$



$\hat{A} \rightarrow$ unit vector

ye aesa vector hai jiski Magnitude '1' aur dirx^n $\hat{A}$ ki taraf hai.

$\rightarrow$ Unit vector along +x-axis = $\hat{i}$

$\rightarrow$ Unit vector along +y-axis = $\hat{j}$

$\rightarrow$ Unit vector along +z-axis = $\hat{k}$

Q. If $\vec{A}$ is unit vector find value of α where $A = 0.6\hat{i} + \alpha\hat{j}$

Sol. $\vec{A} = 0.6\hat{i} + \alpha\hat{j}$

$$|\vec{A}| = \sqrt{0.36 + \alpha^2}$$

$$1 = 0.36 + \alpha^2$$

$$0.64 = \alpha^2$$

$$\pm 0.8 = \alpha$$

Q. If a vector is given by $\vec{A} = 3\hat{i} + 4\hat{j}$. Find unit vector along $\vec{A}$.

Sol. Magnitude of $\vec{A} = \sqrt{3^2 + 4^2} = 5$

$$\vec{A} = |\vec{A}| \cdot \hat{A}$$

dirxⁿ
unit vector
Magnitude

Kisi bhi vector ko uske magnitude se divide kar do to us vector ki taraf ka unit vector aa jaega.



Unit vector along $\vec{A}$ or $\hat{A}$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{3\hat{i} + 4\hat{j}}{5} = \frac{3\hat{i}}{5} + \frac{4\hat{j}}{5}$$

$\vec{A}$ and $\hat{A}$ are parallel vector.

Q. Find the unit vector along $\vec{A}$ where $\vec{A} = 2\hat{i} + 6\hat{j} - 3\hat{k}$

Sol. $\hat{A} = \frac{2\hat{i} + 6\hat{j} - 3\hat{k}}{\sqrt{49}} = \frac{2}{7}\hat{i} + \frac{6}{7}\hat{j} - \frac{3}{7}\hat{k}$

Q. A particle has momentum of magnitude 20 kg m/s. If momentum is in the dirx^n of $\vec{A}$. Find momentum in vector form, where $\vec{A} = \hat{i} + \hat{j}$

Sol. Momentum = $20 \times \hat{A} = 20 \times \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) = \sqrt{2} (10\hat{i} + 10\hat{j})$

Q. A bird is flying with speed 10 m/s in the dirx^n of a vector $\vec{A} = 3\hat{i} + 4\hat{j}$. Find velocity of bird.

Sol. $\vec{V} =$ Magnitude of dirx^n .

$$\vec{V} = 10 \hat{A} = 10 \left[\frac{3\hat{i} + 4\hat{j}}{5} \right] = 2[3\hat{i} + 4\hat{j}] = 6\hat{i} + 8\hat{j}$$

Q. Find force F in vector form if its magnitude is 21 N and dirxⁿ is

- (a) parallel to $6\hat{i} - 3\hat{j} + 2\hat{k}$
 (b) opposite to $6\hat{i} - 3\hat{j} + 2\hat{k}$

Sol. (a) [Parallel] $\vec{F}$

$$= 21\hat{A} = 21 \left[\frac{6\hat{i} - 3\hat{j} + 2\hat{k}}{7} \right] = 18\hat{i} - 9\hat{j} + 6\hat{k}$$

(b) [Opposite] $\vec{F}$

$$= 21(-\hat{A}) = -21\hat{A} = -18\hat{i} + 9\hat{j} - 6\hat{k}$$

Q. A bird is flying with 10 m/s speed from point A to directly point B. Find velocity of Bird.



Sol. $\vec{AB} = 4\hat{i} + 0\hat{j} - 3\hat{k}$

$$\vec{V} = 10 \times \hat{AB} = 10 \times \left(\frac{4\hat{i} - 3\hat{k}}{5} \right) = 8\hat{i} - 6\hat{k}$$

Q. $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$.

$\vec{B} = 2\hat{i} + 3\hat{j} - 2\hat{k}$. Find $\vec{A} + \vec{B}$

Sol. $\vec{A} + \vec{B} = 5\hat{i} + 7\hat{j} + 3\hat{k}$

Q. $\vec{A} = 3\hat{i} + 2\hat{j} + 5\hat{k}$.

$\vec{B} = 2\hat{i} + 3\hat{j} + 4\hat{k}$. Find following vectors.

Sol. $\vec{A} + \vec{B} = 5\hat{i} + 5\hat{j} + 9\hat{k}$

$$\vec{A} - \vec{B} = \hat{i} - \hat{j} + \hat{k}$$

$$2\vec{A} + 3\vec{B} = 12\hat{i} + 13\hat{j} + 22\hat{k}$$

$$|\vec{A} + \vec{B}| = \sqrt{5^2 + 5^2 + 9^2}$$

$$|\vec{A} - \vec{B}| = \sqrt{3}$$

$$|2\vec{A} + 3\vec{B}| = \sqrt{12^2 + 13^2 + 22^2}$$

$$2\vec{A} - 3\vec{B} = 0\hat{i} + (4-9)\hat{j} + (10-12)\hat{k} = -5\hat{j} - 2\hat{k}$$

Q. $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$.

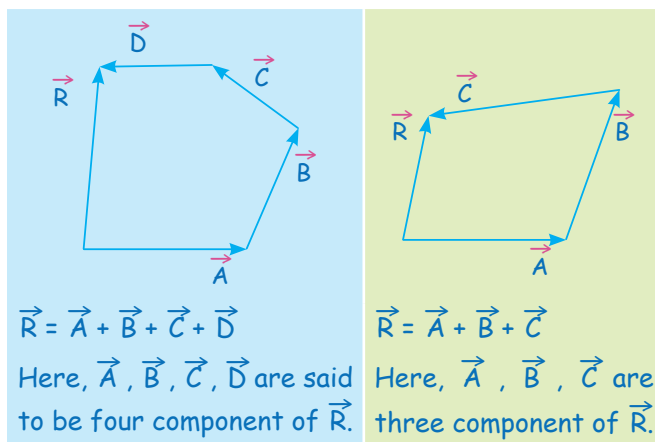
$\vec{B} = 4\hat{i} + \hat{j} + 4\hat{k}$.

Find a vector $\vec{C}$ whose magnitude is 20 and dirxⁿ is opposite to $4\vec{A} + \vec{B}$.

Sol. $4\vec{A} + \vec{B} = 12\hat{i} + 5\hat{j} + 8\hat{k}$

$$\vec{C} = -20 \times \left[\frac{12\hat{i} + 5\hat{j} + 8\hat{k}}{\sqrt{12^2 + 5^2 + 8^2}} \right]$$

COMPONENT OF VECTOR



Q. A man move 5 m along east, then turn left and move 10 m along north, then turn right move 20 m east and then turn right to south & move 15 m. Find net displacement and distance travel.

Sol. Distance = 5 + 10 + 20 + 15 = 50 m

$$\vec{d}_1 = 5\hat{i}$$

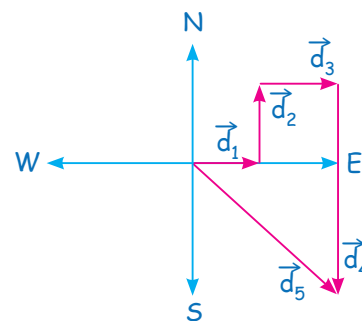
$$\vec{d}_2 = 10\hat{j}$$

$$\vec{d}_3 = 20\hat{i}$$

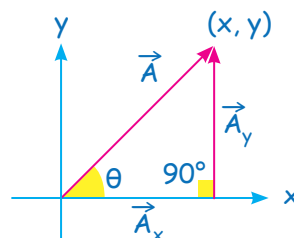
$$\vec{d}_4 = -15\hat{j}$$

$$\vec{d}_{\text{net}} = 25\hat{i} - 5\hat{j}$$

$$\text{Magnitude} = \sqrt{25^2 + 5^2} = \sqrt{625 + 25} = \sqrt{650}$$



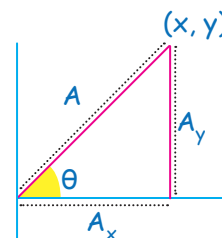
RECTANGULAR COMPONENT/ CARTESIAN COMPONENT OF A VECTOR (2D)



$$\vec{A} = x\hat{i} + y\hat{j},$$

$$\vec{A} = \vec{A}_x + \vec{A}_y,$$

* A_x & A_y are two rectangular component of $\vec{A}$.



$$\vec{A} = A_x \hat{i} + A_y \hat{j},$$

$$\vec{A} = A \cos \theta \hat{i} + A \sin \theta \hat{j},$$

$$\cos \theta = \frac{A_x}{A} \Rightarrow A_x = A \cos \theta$$

$$\sin \theta = \frac{A_y}{A} \Rightarrow A_y = A \sin \theta$$

$\vec{A}_x$ = Component of $\vec{A}$ along x-axis

$\vec{A}_y$ = Component of $\vec{A}$ along y-axis

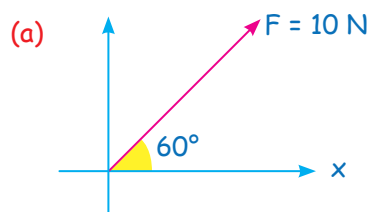
$$\vec{A}_x = A \cos \theta \hat{i}$$

$$\vec{A}_y = A \sin \theta \hat{j}$$

Yeh bahut important hai aur kam se kam ek karod baar use hoga, so, achche se samajh lena ki kaise ek vector ke do rectangular components le rahe hai ya kaise ek vector ko tod rahe hai.

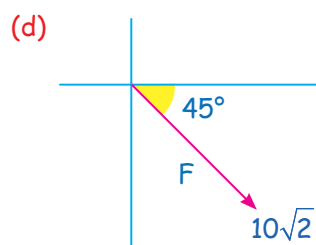
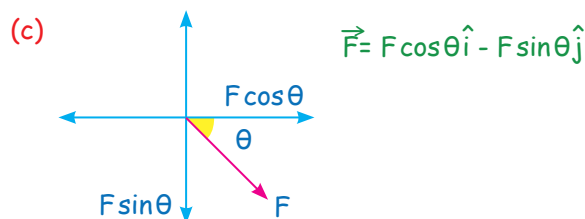
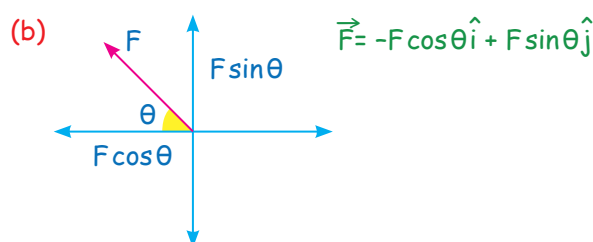


Q. Resolve the vector into $\hat{i}$ & $\hat{j}$

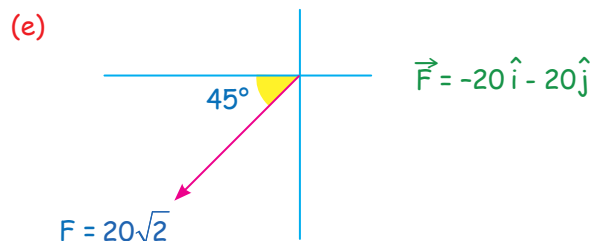


$$\vec{F} = 10 \cos 60^\circ \hat{i} + 10 \sin 60^\circ \hat{j}$$

$$\begin{aligned} \vec{F} &= 10 \times \frac{1}{2} \hat{i} + 10 \times \frac{\sqrt{3}}{2} \hat{j} \\ &= 5\hat{i} + 5\sqrt{3} \hat{j} \end{aligned}$$



$$\vec{F} = 10\sqrt{2} \cos 45^\circ \hat{i} - 10\sqrt{2} \sin 45^\circ \hat{j} = 10\hat{i} - 10\hat{j}$$



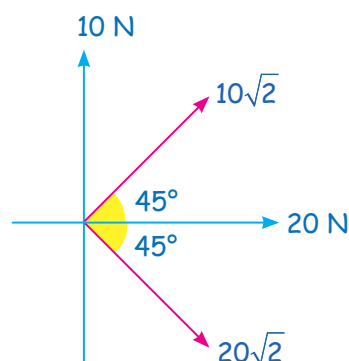
SKC



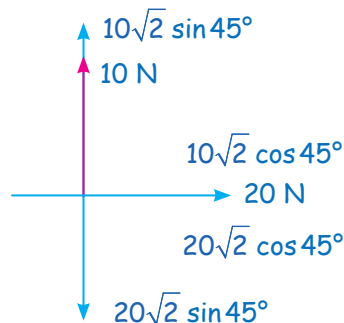
Ab hum vector ka sabse zyada important part padhne jaa rahe hai jo electrostatic, mechanics me bahut use hoga. Yahan humein bahut saare vector ka net resultant nikalna hoga. Kuch nahi karna, bas saare forces ko x-y me tod lo aur individually x aur y me collect karke answer likh do.

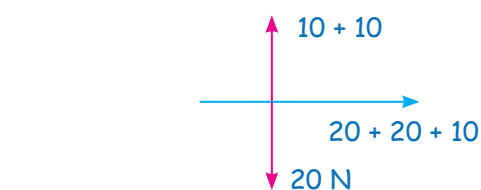


Q. Find net force.



Sol.

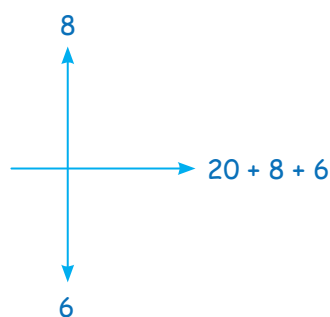
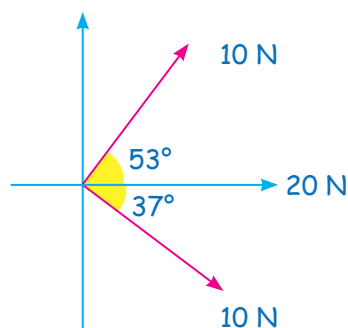




$$F_{\text{net}} = 50\hat{i}$$

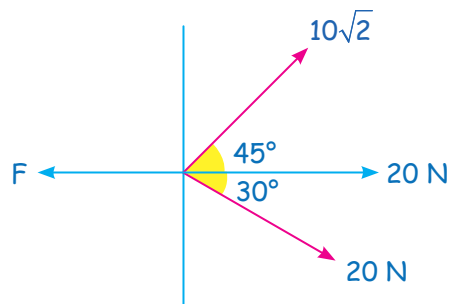
Q. Find net force

Sol.

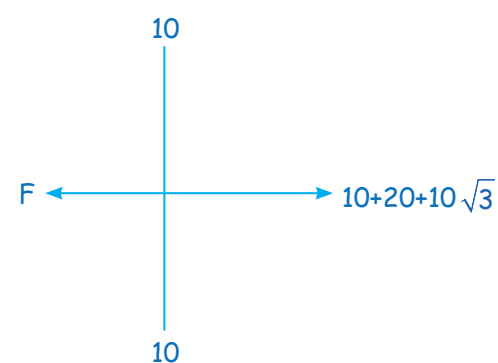


$$F_{\text{net}} = 34\hat{i} + 2\hat{j}$$

Q. Find the value of F so that the net force is zero

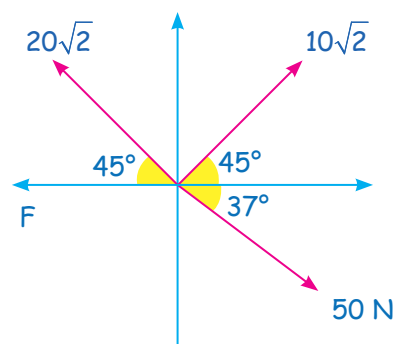


Sol.

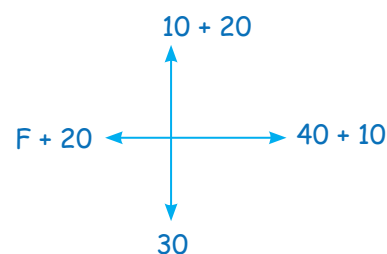


$$30 + 10\sqrt{3} = F$$

Q. Find value of F so that $\vec{F}_{\text{net}} = 0$.



Sol.



$$F + 20 = 50 \Rightarrow F = 30 \text{ (magnitude)}$$

$$\vec{F} = -30\hat{i}$$

Ab agar main 50 vector bhi de doon to sabhi ko resolve karke kar loge na, maanta hoon maths/calculation lambi hogi but physics to aasan hai. Agar ye cheez samajh gaye ho to mujhe insta par confirmation do salim.nitt only if you have account.



VECTOR PRODUCT

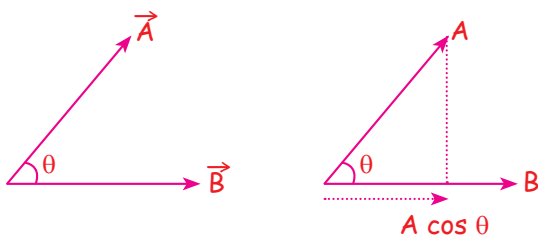
1. Dot Product
2. Cross Product



Dot Product

$$\vec{A} \cdot \vec{B} = A B \cos\theta$$

Here, A and B are the magnitude of $\vec{A}$ and $\vec{B}$ and θ is the angle between $\vec{A}$ and $\vec{B}$.



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = A(B \cos \theta)$$

= A (magnitude of component $\vec{B}$ along $\vec{A}$)

$$\star \vec{A} \cdot \vec{B} = (A \cos \theta)B$$

= (magnitude of component of $\vec{A}$ along $\vec{B}$)B

$$\star \text{ If } \vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = A \cdot B \cos 90^\circ = 0$$

$$\star \hat{i} \cdot \hat{j} = 1 \times 1 \times \cos 90^\circ = 0$$

$$\star \hat{i} \cdot \hat{k} = 1 \times 1 \times \cos 90^\circ = 0$$

$$\star \hat{j} \cdot \hat{k} = 1 \times 1 \times \cos 90^\circ = 0$$

$$\star \vec{A} \cdot \vec{B} = \text{number ayega scalar.}$$

$$\star \hat{i} \cdot \hat{i} = 1 \times 1 \times \cos 0^\circ = 1$$

$$\star \hat{j} \cdot \hat{j} = 1 \times 1 \times \cos 0^\circ = 1$$

$$\star \hat{k} \cdot \hat{k} = 1 \times 1 \times \cos 0^\circ = 1$$

$$\star [\vec{A} \cdot \vec{A} = A^2]$$

$$\text{If } \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\text{then } \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\therefore \vec{A} \cdot \vec{B} = AB \cos \theta \Rightarrow \cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$\star$ Dot product of two vector is commutative

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\star \vec{A} \cdot \vec{A} = A^2$$

$$\star (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = A^2 + B^2 + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A}$$

$$= A^2 + B^2 + 2\vec{A} \cdot \vec{B}$$

Q. Find dot product of $\vec{A}$ and $\vec{B}$.

$$\vec{A} = 2\hat{i} + 3\hat{j}$$

$$\vec{B} = 4\hat{i} + 5\hat{j}$$

$$\vec{A} \cdot \vec{B} = 8 + 15 = 23$$

$$\text{Q. } \vec{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$$

$$\vec{B} = \alpha\hat{i} + 2\hat{j} + 5\hat{k}$$

if $\vec{A} \perp \vec{B}$ find α

$$\text{Sol. } \vec{A} \cdot \vec{B} = 2\alpha - 6 + 20 = 0 \Rightarrow \alpha = -7$$

VVVVVVVVVV Important
agar do vector $\perp$ hai to
unka Dot product zero
hoga

Q. Find angle between $\vec{A}$ & $\vec{B}$ if $\vec{A} = \hat{i} + \hat{j}$, $\vec{B} = 3\hat{i} + 4\hat{j}$

$$\text{Sol. } \vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\text{Magnitude of } B = 5$$

$$\text{Magnitude of } A = \sqrt{2}$$

$$\therefore \cos \theta = \frac{7}{5\sqrt{2}}$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\text{Q. } \vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\vec{B} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

Find angle between $\vec{A}$ & $\vec{B}$

$$\text{Sol. } \vec{A} \cdot \vec{B} = 6 + 12 + 30 = 48, A = 5\sqrt{2}, B = 7$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$48 = 5\sqrt{2} \times 7 \cos \theta$$

$$\frac{48}{35\sqrt{2}} = \cos \theta$$

Q. Find angle between $\vec{A}$ & $\vec{B}$ if

$$\vec{A} = \hat{i} + \hat{j} + \hat{k}, \vec{B} = \hat{i} - \hat{j} + \hat{k}$$

$$\text{Sol. } \vec{A} \cdot \vec{B} = AB \cos \theta$$

$$1 = \sqrt{3} \times \sqrt{3} \cos \theta$$

$$\frac{1}{3} = \cos \theta$$

$$\theta = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\# \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

IF-

$$\frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z} = n$$

• $n > 0 \Rightarrow \vec{A}$ is parallel to $\vec{B}$

• $n < 0 \Rightarrow \vec{A}$ is anti-parallel to $\vec{B}$



$$\text{Q. } \vec{A} = 3\hat{i} - 4\hat{j} + 5\hat{k}$$

$\vec{B} = -9\hat{i} + 12\hat{j} - 15\hat{k}$. Are these vectors parallel or anti parallel?

$$\text{Sol. } \frac{3}{-9} = \frac{-4}{12} = \frac{-5}{15}$$

$$\frac{-1}{3} = \frac{-1}{3} = \frac{-1}{3} \text{ [Antiparallel]}$$

SKC

$\vec{A}, \vec{B} ||^r$ hai ya anti- $||^r$ hai ya $\perp$ hai ya kuch nahi, ye pta karne ke liye

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} \text{ nikal lo.} \Rightarrow \text{If } \theta = 0^\circ = \text{parallel}$$

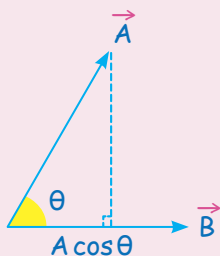
$$\theta = 180^\circ = \text{Anti } ||^r \quad \text{If } \theta = 90^\circ = \perp$$

Q. $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{B} = 4\hat{i} + 6\hat{j} - 8\hat{k}$. Are they parallel?

Sol. $\frac{2}{4} = \frac{3}{6} \neq \frac{-4}{8}$

$$\frac{1}{2} = \frac{1}{2} \neq \frac{-1}{2} \text{ [Nothing]}$$



Component of $\vec{A}$ along $\vec{B} = A \cos \theta$ (magnitude)

$= A \cos \theta \cdot \hat{B}$ (In vector form) $= \vec{A}_{||}$

$$\therefore A \cos \theta = \frac{\vec{A} \cdot \vec{B}}{B}$$

Component of $\vec{A}$ along $\vec{B} = \frac{\vec{A} \cdot \vec{B}}{B} = \vec{A} \cdot \hat{B}$

Component of $\vec{B}$ along $\vec{A} = \frac{\vec{A} \cdot \vec{B}}{A} = \vec{B} \cdot \hat{A}$

Component of $\vec{A}$ along $\vec{B}$ in vector form $(\vec{A}_{||}) = \left(\frac{\vec{A} \cdot \vec{B}}{B} \right) \hat{B}$

Component of $\vec{A}$ perpendicular to $\vec{B} = \vec{A} - \vec{A}_{||}$

$$\vec{A}_{||} + \vec{A}_{\perp} = \vec{A}$$

Iska use aage mechanics mein hoga, isliye rough copy par paanch baar likh-likh kar practice karein (for 11th students).

Q. Find component of $\vec{A}$ along $\vec{B}$

$$\vec{A} = 3\hat{i} + 4\hat{j} \quad \vec{B} = \hat{i} + \hat{j}$$

Sol. Component of $\vec{A}$ along $\vec{B}$ (scalar) $= A \cos \theta$

$$= \frac{\vec{A} \cdot \vec{B}}{B} = \frac{3+4}{\sqrt{2}} = \frac{7}{\sqrt{2}} \text{ (magnitude)}$$

Component of $\vec{A}$ along $\vec{B}$ (Vector) $= \frac{\vec{A} \cdot \vec{B}}{B} \times \hat{B}$

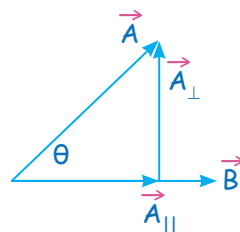
$$= \frac{7}{\sqrt{2}} \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) = \frac{7}{2} (\hat{i} + \hat{j})$$

Q. If $\vec{A} = \hat{i} + 3\hat{j}$ and $\vec{B} = 3\hat{i} + 4\hat{j}$,

Find component of $\vec{A}$ perpendicular to $\vec{B}$.

Sol. Component of $\vec{A}$ parallel (along) to $\vec{B} = \frac{3}{5} (3\hat{i} + 4\hat{j})$

Component of $\vec{A}$ perpendicular to $\vec{B} = A \sin \theta$



$$\vec{A}_{||} + \vec{A}_{\perp} = \vec{A} \Rightarrow \vec{A}_{\perp} = \vec{A} - \vec{A}_{||}$$

$$\vec{A}_{\perp} = \hat{i} + 3\hat{j} - \left[\frac{3}{5} (3\hat{i} + 4\hat{j}) \right] = \hat{i} + 3\hat{j} - \left(\frac{9\hat{i} + 12\hat{j}}{5} \right)$$

$$= \frac{-4\hat{i}}{5} + \frac{3\hat{j}}{5}$$

Q. $\vec{A} = 4\hat{i} - 2\hat{j}$, $\vec{B} = 3\hat{i} + 4\hat{j}$

Find component of $\vec{A}$ perpendicular to $\vec{B}$.

Sol. Component of $\vec{A}$ parallel to $\vec{B}$

$$= \frac{\vec{A} \cdot \vec{B}}{B} = \frac{12-8}{5} = \frac{4}{5}$$

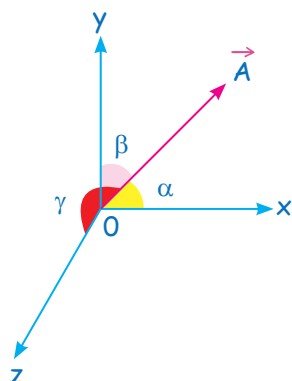
$$\text{Vector form} = \frac{4}{5} \left(\frac{3\hat{i} + 4\hat{j}}{5} \right) = \frac{12\hat{i} + 16\hat{j}}{25}$$

Component of $\vec{A}$ perpendicular to $\vec{B}$

$$= \vec{A}_{\perp} = \vec{A} - \vec{A}_{||}$$

$$= 4\hat{i} - 2\hat{j} - \left[\frac{12\hat{i} + 16\hat{j}}{25} \right] = \frac{88\hat{i} - 66\hat{j}}{25}$$

DIRECTION COSINE



$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

If $\vec{A}$ makes angle α with + x-axis

If $\vec{A}$ makes angle β with + y-axis

If $\vec{A}$ makes angle γ with + z-axis

Component of $\vec{A}$ along x-axis = $A \cos \alpha = A_x$

Component of $\vec{A}$ along y-axis = $A \cos \beta = A_y$

Component of $\vec{A}$ along z-axis = $A \cos \gamma = A_z$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

$$\cos \alpha = \frac{A_x}{A} \quad \cos \beta = \frac{A_y}{A} \quad \cos \gamma = \frac{A_z}{A} \quad \text{here, } \cos \alpha, \cos \beta,$$

$\cos \gamma$ are the direction cosine of $\vec{A}$

Q. $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$. Find direction cosine

Sol. $A = |\vec{A}| = \sqrt{2^2 + 3^2 + 6^2} = 7$

$$\cos \alpha = \frac{A_x}{A} = \frac{2}{7} \quad \cos \beta = \frac{A_y}{A} = \frac{3}{7} \quad \cos \gamma = \frac{A_z}{A} = \frac{6}{7}$$

Q. $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$, Find direction cosine

Sol. $A = \sqrt{9 + 16 + 25} = 5\sqrt{2}$

(a) Direction cosine

$$\cos \alpha = \frac{3}{5\sqrt{2}} \quad \cos \beta = \frac{4}{5\sqrt{2}} \quad \cos \gamma = \frac{5}{5\sqrt{2}}$$

(b) Component of $\vec{A}$ along x-axis = $3\hat{i}$

(c) Component of $\vec{A}$ along y-axis = $4\hat{j}$

(d) Component of $\vec{A}$ along z-axis = $5\hat{k}$

(e) Component of $\vec{A}$ on x-y plane = $3\hat{i} + 4\hat{j}$

(f) Component of $\vec{A}$ on y-z plane = $4\hat{j} + 5\hat{k}$

Cross Product

Bhai cross product ki zyada tension mat lena bas ye sikh lena ki kaise cross product nikalte hain. Rotation se pehle iska bahut hi kam use hoga, thoda sa circular motion mein use hoga.

If cross product of vector $\vec{A}$ and $\vec{B}$ is $\vec{C}$ means $\vec{A} \times \vec{B} = \vec{C}$ then $\vec{C}$ is a vector which is perpendicular to both $\vec{A}$ and $\vec{B}$ and magnitude of $\vec{C}$ is given by

$$|\vec{C}| = AB \sin \theta$$

$$\star \vec{C} = (AB \sin \theta) \hat{C}$$

$$\star \vec{a} \times \vec{b} = \vec{c} = ab \sin \theta \hat{c}$$

$$\star \vec{c} \cdot \vec{a} = 0, \vec{c} \cdot \vec{b} = 0$$

$$\star \vec{c} \perp \vec{a}, \vec{c} \perp \vec{b}$$

$$\star (\vec{a} \times \vec{b}) \cdot \vec{a} = 0, (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

$$\star \vec{a} \times \vec{a} = 0$$

$$\star \hat{i} \times \hat{i} = 0, \hat{j} \times \hat{j} = 0, \hat{k} \times \hat{k} = 0$$

2 vector ka cross product khud me ek vector hota hai.

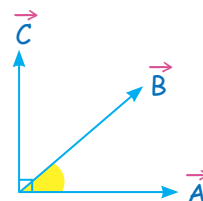


$$\star \text{ In terms of components } \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_z B_x - A_x B_z) + \hat{k}(A_x B_y - A_y B_x)$$

Agar aapne determinant nahi padha, to tension lene ki koi zaroorat nahi hai. Jab padh lo tab yeh method apply karke cross product nikal lena.

$$\star \vec{A} \times \vec{B} = \vec{C}$$



SKC

$$\begin{aligned} \star \hat{i} \times \hat{j} &= \hat{k} \\ \star \hat{j} \times \hat{k} &= \hat{i} \\ \star \hat{k} \times \hat{i} &= \hat{j} \\ \star \hat{j} \times \hat{i} &= -\hat{k} \\ \star \hat{i} \times \hat{k} &= -\hat{j} \\ \star \hat{k} \times \hat{j} &= -\hat{i} \end{aligned}$$



Ab yaad karne ka tareeka sun, zubaan se bolo $\hat{i} \hat{j} \hat{k} \hat{i} \hat{j} \hat{k} \hat{i} \hat{j} \hat{k} \hat{i} \hat{j} \hat{k}$ aur ye sequence catch karo jaise i aur j ke baad k aaya to $\hat{i} \times \hat{j} = \hat{k}$ usi sequence mein j aur k ke baad i aaya hai to $\hat{j} \times \hat{k} = \hat{i}$ agar sequence gadbad/ulta hua to minus laga do jaise $\hat{j} \times \hat{i} = -\hat{k}$ abe, samajh gaye na.



Q. $\vec{a} = 3\hat{i} + 4\hat{j}$

$\vec{b} = 2\hat{i} + 5\hat{j}$ find $\vec{a} \times \vec{b}$

Sol. $\vec{a} \times \vec{b} = (3\hat{i} + 4\hat{j}) \times (2\hat{i} + 5\hat{j})$
 $= 6\hat{i} \times \hat{i} + 15\hat{i} \times \hat{j} + 8\hat{j} \times \hat{i} + 20\hat{j} \times \hat{j}$
 $= 0 + 15\hat{k} - 8\hat{k} + 0$
 $= 7\hat{k} = \vec{c}$
 $\vec{c} \perp \vec{a}$
 $\vec{c} \perp \vec{b}$

Q. $\vec{a} = 4\hat{i} + 7\hat{j}$

$\vec{b} = 2\hat{i} + 3\hat{j}$ find $\vec{a} \times \vec{b}$

Sol. $\vec{a} \times \vec{b} = 8\hat{i} \times \hat{i} + 12\hat{i} \times \hat{j} + 14\hat{j} \times \hat{i} + 21\hat{j} \times \hat{j}$
 $= 0 + 12\hat{k} - 14\hat{k} + 0 = -2\hat{k} = \vec{c}$
 $\vec{c} \perp \vec{a}, \vec{c} \perp \vec{b}$
 $\vec{b} \times \vec{a} = (2\hat{i} + 3\hat{j}) \times (4\hat{i} + 7\hat{j})$
 $= 0 + 14\hat{k} - 12\hat{k} + 0 = 2\hat{k}$
 $\therefore \vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$

Ab main neeche kuch important question attach kar raha hoon, inko achche se solve karo... Abe karoge na.....



KUCHH PRACTICE KRLO

Q. If the sum of two unit vectors is a unit vector, then find the magnitude of their difference.

Sol. Let $\hat{n}_1$ and $\hat{n}_2$ are the two unit vectors, then the sum is $\vec{n}_s = \hat{n}_1 + \hat{n}_2$ or $n_s^2 = n_1^2 + n_2^2 + 2n_1n_2 \cos \theta = 1 + 1 + 2 \cos \theta$

Since it is given that n_s is also a unit vector, therefore $1 = 1 + 1 + 2 \cos \theta$

$$\text{or } \cos \theta = -\frac{1}{2} \text{ or } \theta = 120^\circ$$

Now the difference of the vectors is $\vec{n}_d = \hat{n}_1 - \hat{n}_2$

$$\text{or } n_d^2 = n_1^2 + n_2^2 - 2n_1n_2 \cos \theta = 1 + 1 - 2 \cos(120^\circ)$$

$$\therefore n_d^2 = 2 - 2(-1/2) = 2 + 1 = 3 \Rightarrow n_d = \sqrt{3}$$

Q. If vector $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = -4\hat{i} - 6\hat{j} - \lambda\hat{k}$ are perpendicular to each other then value of λ will be?

Sol. If $\vec{A}$ and $\vec{B}$ are perpendicular to each other then $\vec{A} \cdot \vec{B} = 0$

$$\text{So, } 2(-4) + 3(-6) + (-1)(-\lambda) = 0 \Rightarrow \lambda = +26$$

Q. If $\vec{a}_1$ and $\vec{a}_2$ are two non collinear unit vectors and if $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then find the value of $(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2)$.

Sol. $a_1 = a_2 = 1$ and $a_1^2 + a_2^2 + 2a_1a_2 \cos \theta = 3$

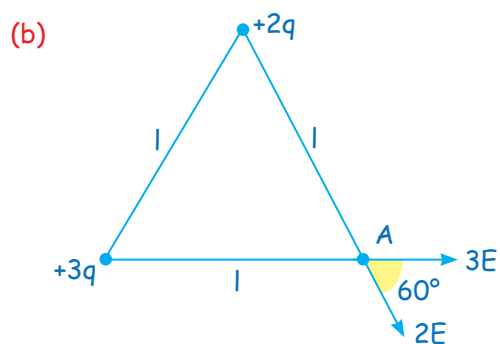
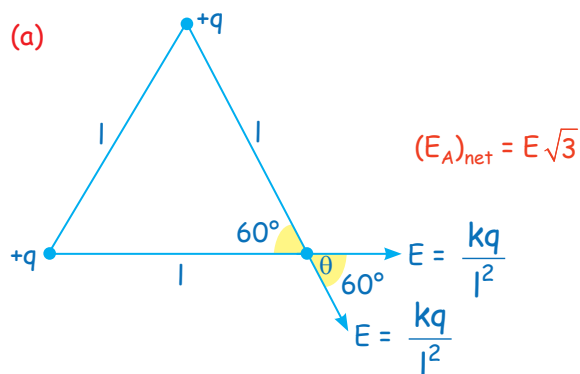
$$\text{Or } 1 + 1 + 2 \cos \theta = 3 \text{ or } \cos \theta = \frac{1}{2}$$

$$\text{Now } (\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2) = 2a_1^2 - a_2^2 - a_1a_2 \cos \theta$$

$$= 2 - 1 - \frac{1}{2} = \frac{1}{2}$$

Q. In 12th class we will study in electrostatics that a point charge produce electric field which is a vector quantity and electric field due to a positive point charge at a distance r from charge has value $\frac{kq}{r^2}$ radially away from charge (along the line joining point to the charge) k is constant

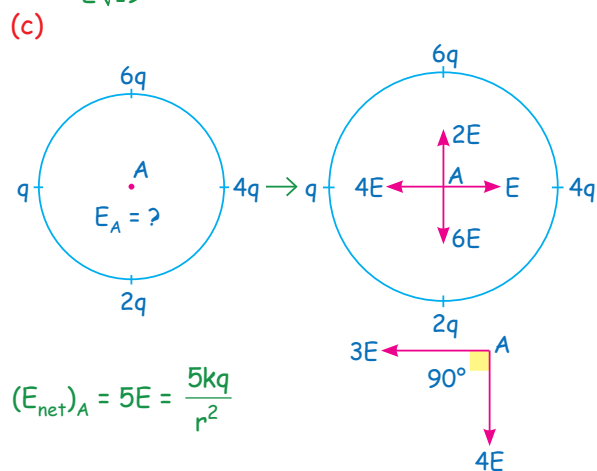
Using above data find the net electric field at point A in following case.



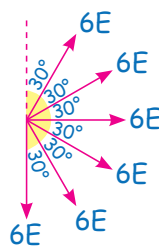
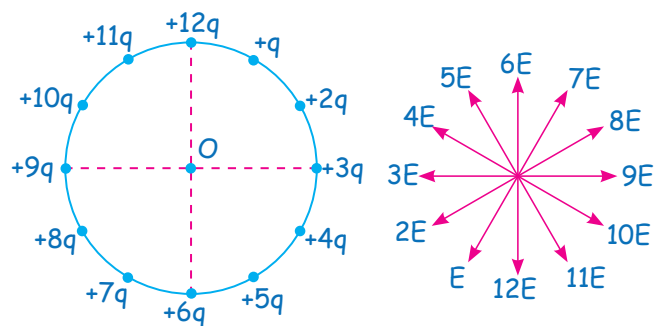
Let $E = \frac{kq}{l^2}$

$$E_{\text{net}} = \sqrt{(2E)^2 + (3E)^2 + 2 \times 2E \times 3E \times \cos 60^\circ}$$

$$= E\sqrt{19}$$



(d) Find E_{net} at centre O.



Ab component lekar solve kar loge na... Nahi to koi baat nahi 12th mein kar lena.

$$\vec{E}_{\text{net}} = (12E + 6E\sqrt{3})\hat{i} - 6E\hat{j}$$

IMPORTANT RESULT

If $|\vec{A}| = |\vec{B}| = x$

and angle between $\vec{A}$ and $\vec{B}$ is θ

then $|\vec{A} + \vec{B}| = 2x \cos \theta/2$

$|\vec{A} - \vec{B}| = 2x \sin \theta/2$

Q. Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P} + \vec{Q}$ is n times the magnitude of $\vec{P} - \vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is:

Ans. $\cos^{-1}\left(\frac{n^2 - 1}{n^2 + 1}\right)$